Chenmu Zhang

Houston, TX
Mobile: (+1) 737-228-8750 Email: zcmben2014@gmail.com
chenmuzhang.com Google Scholar github.com/cz2014

RESEARCH INTERESTS

- **Agentic AI for computational materials science:** autonomous LLM agents for simulation and materials discovery; agent-driven model development; benchmarks and evaluation of agents on research tasks
- **Multi-agent autonomous research:** parallel exploration and knowledge sharing among agent populations; organization of AI research groups; scaling of discovery with the number of agents
- **Machine learning for electronic structure and dynamics:** learned Hamiltonians and interatomic potentials; surrogates for first-principles dynamics

EDUCATION

► **University of Texas at Austin, Austin, TX** Aug. 2018 – Apr. 2023

Ph.D. in Materials Engineering
Advisor: Prof. Yuanyue Liu
Thesis: *"Phonon-limited Carrier Transport in 2D Materials from First Principles"*

► **University of Science and Technology of China (USTC), Hefei, China** Sep. 2014 – Jul. 2018

B.S. in Physics
Advisor: Prof. Wenguang Zhu
Thesis: *"First Principles Calculations on 2D Ferroelectric Materials under Uniaxial Strain"*

RESEARCH EXPERIENCE

► **Postdoctoral Research Associate** May 2023 – Present

Advisor: Prof. Boris I. Yakobson, Rice University, Houston, TX

- Designed and built MatClaw, an open-source code-first LLM agent that carries out long-running first-principles workflows on HPC clusters (arXiv 2604.02688).
- Applied MatClaw to a materials-discovery study of off-stoichiometric Cu–In–Se, where the agent ran generative structure prediction, interatomic-potential screening and DFT validation end to end (co-first-author manuscript, in preparation).
- Applied autonomous research to neural-network design for band-gap prediction: an LLM agent redesigned, trained and evaluated the model over 184 experiments with no human input; the final model ranks first among from-scratch models on the public MatBench leaderboard, ahead of all 17 expert-designed ones (arXiv 2606.29717; LM4Sci workshop, COLM 2026).
- Designed CompMat-Bench, a benchmark of 94 tasks derived from recently published computational materials studies, with rule-based grading against reproduced results; evaluated four agents on three LLMs under single-task and workflow conditions (arXiv 2610.00636, under review).
- Defect physics as a design lever: stochastic defect dynamics in 2D semiconductors as an entropy source for true random bits; mechanisms of defect-governed heat transport at extreme temperatures (two manuscripts submitted).

► **Graduate Research Assistant** Aug. 2018 – Apr. 2023

Advisor: Prof. Yuanyue Liu, University of Texas at Austin, Austin, TX

- Established why 2D semiconductors generally have low mobility and which electronic structures escape it, guiding the discovery of room-temperature high-mobility 2D semiconductors (PRL 2020, 2023).
- Extended first-principles electron-phonon transport to quadrupole, substrate, surface, and remote-phonon scattering, revealing anomalous mobility enhancement by remote phonons (ACS Nano 2024; Nano Lett. 2026); built a computational carrier-mobility genome for 2D semiconductors.

PUBLICATIONS

13 peer-reviewed publications (8 first-author), including *Phys. Rev. Lett.*, *J. Am. Chem. Soc.*, *ACS Nano*, and *Nano Lett.*; 3 manuscripts under review; 613 citations, h-index 10 (Google Scholar, Aug. 2026).

19. **Zhang, C.**; Felix, L. C.; Zhang, J.-J.; Li, X.; Jiang, X.; Jiang, T.; Mishra, S.; Qin, X.; Yakobson, B. I. CompMat-Bench: Benchmarking AI Agents for Computational Materials Science. arXiv preprint arXiv:2610.00636, (Submitted) **2026**.
18. Yuan*, Z.; **Zhang*, C.** Off-Stoichiometric Energetic and Electronic Asymmetry in Ternary Cu–In–Se: Task-Oriented Materials Discovery by an Autonomous LLM Agent. (In preparation; * equal contribution) **2026**.
17. **Zhang*, C.**; Felix*, L. C.; Yakobson, B. I. Defect-controlled thermal conductivity of ZrC at high temperatures from first principles. (Submitted; * equal contribution) **2026**.
16. **Zhang, C.**; Sun, F.; Yakobson, B. I. Design of efficient true random bits generation in transition metal dichalcogenides from first principles. (Submitted) **2026**.
15. **Zhang, C.**; Yakobson, B. I. Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. arXiv preprint arXiv:2606.29717, accepted at the Workshop on Language Models for Scientific Discovery (LM4Sci), COLM **2026**.
14. **Zhang, C.**; Liu, Y. Enhancing Carrier Mobility by Remote Phonons. *Nano Letters* **2026**, 26(30), 9716–9721.
13. **Zhang, C.**; Yakobson, B. I. MatClaw: An Autonomous Code-First LLM Agent for End-to-End Materials Exploration. arXiv preprint arXiv:2604.02688, **2026**.
12. **Zhang, C.**; Liu, Y. Two-dimensional Semiconductor Computational Carrier Mobility Genome. arXiv preprint arXiv:2404.08117, **2024**.
11. **Zhang, C.**; Liu, Y. Electron-Surface Scattering from First-Principles. *ACS Nano* **2024**, 18(40), 27433–27439.
10. **Zhang, C.**; Xiao, Z.; Paddock, R.; Cullinan, M.; Tehrani, M.; Liu, Y. Effects of Graphene Doping on the Electrical Conductivity of Copper. *Advanced Functional Materials* **2024**, 34(45), 2407569.
9. **Zhang, C.**; Wang, R.; Mishra, H.; Liu, Y. Two-Dimensional Semiconductors with High Intrinsic Carrier Mobility at Room Temperature. *Physical Review Letters* **2023**, 130(8), 087001.
8. **Zhang, C.**; Liu, Y. Phonon-limited transport of two-dimensional semiconductors: Quadrupole scattering and free carrier screening. *Physical Review B* **2022**, 106(11), 115423.
7. **Zhang, C.**; Cheng, L.; Liu, Y. Understanding high-field electron transport properties and strain effects of monolayer transition metal dichalcogenides. *Physical Review B* **2020**, 102(11), 115405.
6. **Zhang, C.**; Cheng, L.; Liu, Y. Role of flexural phonons in carrier mobility of two-dimensional semiconductors: free standing vs on substrate. *Journal of Physics: Condensed Matter* **2021**, 33(23), 234003.
5. Xiao, Z.; Guo, R.; **Zhang, C.**; Liu, Y. Point Defect Limited Carrier Mobility in 2D Transition Metal Dichalcogenides. *ACS Nano* **2024**, 18(11), 8511–8516.
4. Cheng, L.; **Zhang, C.**; Liu, Y. Intrinsic charge carrier mobility of 2D semiconductors. *Computational Materials Science* **2021**, 194, 110468.
3. Cheng, L.; **Zhang, C.**; Liu, Y. Why Two-Dimensional Semiconductors Generally Have Low Electron Mobility. *Physical Review Letters* **2020**, 125(17), 177701.
2. Cheng, L.; **Zhang, C.**; Liu, Y. The Optimal Electronic Structure for High-Mobility 2D Semiconductors: Exceptionally High Hole Mobility in 2D Antimony. *Journal of the American Chemical Society* **2019**, 141(41), 16296–16302.
1. Cheng, L.; **Zhang, C.**; Liu, Y. How to resolve a phonon-associated property into contributions of basic phonon modes. *Journal of Physics: Materials* **2019**, 2(4), 045005.

SOFTWARE

- **MatClaw** — code-first LLM agent for end-to-end first-principles materials research: persistent Python kernels, self-written tools, supervised runs on remote HPC clusters. Open source; sole developer.
- **CompMat-Bench** — benchmark of AI agents on tasks drawn from recently published computational materials studies; rule-based grading, no LLM judge. Code release upon publication.
- **qeh-frohlich** — phonon-limited mobility in van der Waals heterostructures with Fröhlich coupling. Sole developer.
- **EDI** — first-principles electron-defect scattering and defect-limited transport. Contributor.

PRESENTATIONS

- *How Well Can AI Agents Do Computational Materials Research? A Benchmark Built from Peer-Reviewed Publications*, MRS Fall Meeting, Symposium MT01, Boston, Dec. 2026 (talk, scheduled)
- *Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop*, LM4Sci Workshop @ COLM 2026, San Francisco, Oct. 2026 (poster, scheduled)
- *Autonomous Materials Simulation with an LLM Agent: Modeling Exothermic Fronts in 2D Ferroelectrics*, APS Global Physics Summit, Denver, 2026 (talk)
- *First-Principles Electron-Surface Scattering: Case Study on Copper Films*, APS Global Physics Summit, Anaheim, 2025 (talk); *Harnessing Stochasticity in 2D Materials: Phonon-Assisted Defect Charge Dynamics in WS_2*, same meeting (poster)
- *Electron-Surface Scattering from First Principles*, APS March Meeting, Minneapolis, 2024 (talk); *New 2D Semiconductors with Ultra-High Mobility at Room Temperature*, same meeting (poster)

PROFESSIONAL SERVICE

- Reviewer for *Physical Review B*, *Ceramics International*, and *Journal of Superconductivity and Novel Magnetism*

TEACHING

- Teaching Assistant, Structure of Materials (Fall 2021, Fall 2022), University of Texas at Austin